

- 1) The purpose of a loop is to repeat a set of steps until something changes or the condition is no longer true.
- 2) A while loop checks the condition before it runs, so it might not run at all. A do-while loop runs once before checking, so it always runs at least once.
- 3) The input validation loop keeps asking the user for input until they enter something valid.
- 4) a) An infinite loop is a loop that never stops.
b) It can happen if you forget to change the variable that controls the loop or if the condition never becomes false.
c) Overflow happens when a number gets too big and wraps around to a smaller or negative number.
- 5) The loop runs 60 times because it keeps adding 2 to x until it reaches 120.
- 6) If x starts at 120 or less and keeps going down by 3, it will never reach 120, so the loop never ends.
- 7) A counter counts how many times something happens (like button clicks).
An accumulator adds up numbers to get a total (like adding prices).
Counters usually go up by 1, and accumulators can increase by different amounts.
- 8) `sum = 0`
`i = 3 to 10`
`sum = sum + i`
→ This adds all numbers from 3 to 10.
- 9) If you know exactly how many times the loop should run, use a for loop.
If the loop needs to run at least once, use a do-while loop.
- 10) `x = "my string."`
a) `x.length` → 10 (there are 10 characters including the space and period)
b) `x.substring(0, 3)` → "my " (it takes the first three characters)